

PURPLE TEAM HANDOVER REPORT

HawkEye Reborn v9 Attack Analysis and Defensive Improvement Report

Report	Task 3 - Purple Team Handover Report
Analysis	HawkEye Reborn v9 Attack Analysis and Defensive Improvement Report
Incident Type	Credential Theft and Data Exfiltration
Malware Family	HawkEye Reborn v9

Executive Summary

This report presents a combined offensive and defensive analysis of the HawkEye Reborn v9 malware incident. The objective is to provide a complete understanding of the attack lifecycle, the techniques used by the attacker, the investigation process conducted by defenders, and recommendations for improving the organization's security posture.

The investigation confirmed that HawkEye Reborn v9 successfully infected a workstation, harvested sensitive credentials, and exfiltrated the collected data through SMTP communications to attacker-controlled infrastructure.

1. Incident Overview

Field	Value
Incident Type	Credential Theft and Data Exfiltration
Malware Family	HawkEye Reborn v9
Affected Host	BEIJING-5CD1-PC
Internal IP Address	10.4.10.132
Public IP Address	173.66.146.112
Victim User	roman.mcguire
Attack Objective	Steal user credentials and transmit them to an external attacker.

2. Red Team Findings (Attack Perspective)

The attack began when the victim executed a malicious file named tkraw_Protected99.exe.

Following execution, HawkEye Reborn v9 performed the following actions:

- Collected system information.
- Gathered hostname and network information.
- Harvested browser-stored credentials.
- Captured email account credentials.
- Stole banking credentials.
- Established communication with external infrastructure.
- Sent stolen information through SMTP.

Attacker Infrastructure

Indicator	Value
SMTP Server	23.229.162.69
SMTP Hostname	p3plcpnl0413.prod.phx3.secureserver.net
Email Account	sales.del@macwinlogistics.in
Exfiltration Frequency	Every 10 minutes

3. Blue Team Findings (Defensive Perspective)

Network traffic analysis identified suspicious SMTP activity originating from the victim workstation.

Investigators performed:

- Packet capture analysis using Wireshark.
- DNS investigation.
- HTTP object extraction.
- SMTP stream reconstruction.
- Base64 decoding of SMTP credentials.
- Malware attribution.

Recovered Evidence

Evidence	Value
Malicious File	tkraw_Protected99.exe
Malicious Domain	proforma-invoices.com
Resolved Address	217.182.138.150
Web Server	LiteSpeed

Recovered Credentials

- AOL Account
- Outlook Account
- Bank of America Account
- Bank of America Credentials: roman.mcguire:P@ssw0rd\$

4. MITRE ATT&CK; Mapping

Tactic	Technique ID	Technique
Execution	T1204.002	User Execution: Malicious File
Credential Access	T1056.001	Keylogging
Credential Access	T1555	Credentials from Password Stores
Discovery	T1082	System Information Discovery
Collection	T1119	Automated Collection
Exfiltration	T1048.003	Exfiltration Over SMTP

5. Security Gaps Identified

The investigation revealed several weaknesses:

- Users were able to execute untrusted executable files.
- Endpoint protection did not prevent malware execution.
- SMTP traffic from workstations was not restricted.
- Sensitive credentials were stored in accessible locations.
- Multi-Factor Authentication was not enforced.

6. Detection Opportunities

Email Security

- Block executable attachments.
- Scan incoming attachments for malware.

Endpoint Monitoring

- Detect execution of suspicious executable files.
- Monitor credential dumping and keylogging behavior.

Network Monitoring

- Alert on outbound SMTP traffic from user workstations.
- Detect communication with known malicious domains.
- Monitor repeated data transfers to external email servers.

Threat Hunting

- Search for SMTP authentication attempts from endpoints.
- Investigate unusual DNS requests and external connections.

7. Recommendations

Immediate Actions

- Reset compromised credentials.
- Block identified malicious domains and IP addresses.
- Remove infected systems from the network.

Long-Term Improvements

- Deploy Endpoint Detection and Response (EDR).
- Enforce Multi-Factor Authentication (MFA).
- Implement Security Information and Event Management (SIEM) monitoring.
- Conduct employee phishing and malware awareness training.
- Restrict outbound SMTP traffic to approved mail servers only.

8. Lessons Learned

The incident demonstrated how information-stealing malware can abuse legitimate protocols such as SMTP to bypass traditional security controls. Effective defense requires a combination of endpoint protection, network monitoring, user awareness, and proactive threat hunting.

9. Conclusion

The HawkEye Reborn v9 incident highlighted the complete lifecycle of a credential theft attack. The Red Team analysis demonstrated how the malware successfully infected the victim and exfiltrated sensitive information, while the Blue Team investigation identified attacker infrastructure, recovered stolen credentials, and documented indicators of compromise. The findings from both teams provide valuable insights for strengthening organizational defenses and improving future detection and response capabilities.